

SANJAY CH

DATA ENGINEER

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PROFESSIONAL SUMMARY

Data Engineer with nearly 5 years designing and automating data workflows on **AWS** and **Azure**, processing **5M+ records and 2 TB+** data daily through **ETL pipelines** in **Databricks** and **Snowflake** across healthcare and financial domains. Proficient in **Python** and **SQL** for optimizing queries and database performance on **Redshift** and **SQL Server**, improving reporting speed by 55% while lowering compute costs by 30%. Led end-to-end pipeline design and collaborated with **cross-functional teams** (product, BI, and data science) to achieve 99% accuracy and deliver real-time insights for claims and financial operations. Automated validation and CI/CD deployments using **Terraform**, **Jenkins**, and **Git**, cutting release cycles by 40% and maintaining **HIPAA/SOX** compliance. Integrated **GenAI-based document summarization** and **ML anomaly-detection models** to enhance fraud monitoring and accelerate decision-making.

TECHNICAL SKILLS

Machine Learning & AI:	Gen AI, LLM's, TensorFlow, Scikit-learn, MLflow, Applied ML Pipelines
Programming & Frameworks:	Python, SQL, PySpark, Java, Scala
Data Engineering & Cloud:	Azure (Data Factory, Databricks, Synapse, ADLS Gen2, Purview, Key Vault), AWS (Glue, Redshift, S3, Lambda, Bedrock), Snowflake, Apache Spark
Database & Storage:	SQL Server, PostgreSQL, MySQL, Oracle, MongoDB, Delta Lake
Orchestration & Monitoring:	Airflow, Azure Data Factory, Apache Kafka, dbt, Great Expectations, Azure Monitor
DevOps & Infrastructure:	Azure DevOps, Jenkins, Git, Terraform, Docker
Analytics & Visualization:	Power BI, Amazon QuickSight, Tableau, Advanced Excel (Pivot, VLOOKUP)
Governance & Security:	Azure Purview, Azure Active Directory (AAD), Data Quality, Compliance (HIPAA, SOX)

CERTIFICATIONS

- AWS Certified Solutions Architect – Professional
- Microsoft Certified: Azure AI Engineer Associate

PROFESSIONAL EXPERIENCE

Data Engineer | **Blue Cross Blue Shield of Texas, TX** May 2024– Present

Blue Cross Blue Shield, a leading U.S. health insurer, was constrained by manual claim workflows and inconsistent data validation. Automated end-to-end processing pipelines that accelerated approvals by 35 %, improved claims accuracy, and reduced escalations across operations.

- Engineered **GenAI services** powered by **LLMs** to extract policy clauses, medical codes, and claim details from unstructured healthcare documents, improving claims accuracy by 35% and automating 40% of manual review tasks.
- Improved compliance alignment and system transparency across engineering and clinical teams by designing healthcare data architecture with **Data Flow Diagrams (DFDs)** to map claims, **EHR**, and lab ingestion pipelines.
- Architected **distributed ETL systems** with **AWS Glue** and **PySpark**, integrating genomic, imaging, and lab records into **Amazon Redshift**, enabling real-time population health analytics.
- Reduced downtime by 55% and manual interventions by 45% through automated claims validation and record reconciliation using **Apache Airflow**, **AWS Lambda**, and **Step Functions** with error-tolerant recovery.
- Enhanced patient risk scoring accuracy by 30% by developing **TensorFlow** ML models predicting chronic disease progression (diabetes, heart failure) with advanced feature engineering and cross-validation.
- Cut reporting latency across hospital and clinic networks by 35% by leveraging **Apache Hive & HiveQL** on **AWS EMR** to query large-scale **EHR & claims data**, optimizing aggregation of healthcare quality measures.
- Implemented **CI/CD pipelines** with **Jenkins** and **AWS CodePipeline**, automating ETL and ML deployments with version control, automated testing, and HIPAA-compliant release workflows.
- Accelerated access to eligibility, prescription, and claims data by 55% through optimization of **MySQL** databases and migration of workloads using **AWS DMS**, reducing query response times significantly.
- Developed **interactive analytics applications** in **Power BI** and **Amazon QuickSight**, delivering real-time insights into admission trends, treatment outcomes, and provider performance for leadership teams.
- Strengthened PHI security and interoperability by enforcing **HIPAA** and **HL7 compliance** with **AWS Lake Formation** and **CloudWatch**, implementing encryption, governance, and role-based access controls.

- Collaborated in **Agile** sprints with **cross-functional teams**, delivering 95% of planned features on time and improving release velocity by 20% through iterative development and continuous feedback.

Data Engineer | Capgemini, India

Aug 2020 – April 2023

Capgemini, a global consulting leader serving major banking clients, faced reporting latency and compliance bottlenecks.

Engineered connected data workflows that enhanced fraud detection by 48 %, accelerated regulatory reporting by 30 %, and improved executive decision speed.

- Migrated on-prem Hadoop workloads to **Azure Data Lake Storage Gen2** using **Azure Data Factory** and **Databricks (PySpark)**, modernizing financial-data infrastructure and cutting processing latency by 40%.
- Engineered ETL pipelines in **Azure Databricks (PySpark + SQL)** to consolidate mortgage, transaction, and securities data into **Azure Synapse Analytics**, improving query performance by 35%.
- Optimized Spark jobs through broadcast joins, partition tuning, and Parquet conversion within **Databricks**, reducing runtime from 80 to 25 minutes and lowering compute cost by 25%.
- Implemented data-validation frameworks with **Python**, **T-SQL**, and **Great Expectations**, automating reconciliation across **SQL Server** sources and sustaining 99% accuracy for audit-ready reports.
- Standardized governance and security controls via **Azure Purview** and **Key Vault**, ensuring SOX compliance and secure credential management across pipelines.
- Operationalized fraud-risk and credit-scoring models on **Azure Machine Learning** with **MLflow** on **Databricks**, enabling batch and streaming scoring via **Event Hubs** and reducing false positives by 22% while improving alert lead time by 30%.
- Collaborated with backend teams to design lightweight **Java Spring Boot** APIs exposing curated financial datasets for internal dashboards, improving data-access efficiency by 20%.
- Automated CI/CD deployments using **Azure DevOps** and **Git**, reducing release time by 30% and eliminating manual promotion errors.
- Containerized data-processing environments with **Docker** and managed infrastructure through **Terraform**, ensuring consistent builds and accelerating environment setup by 45%.
- Partnered with finance and analytics teams under **Agile** sprints to deliver governance dashboards in **Power BI**, strengthening transparency and reporting accuracy.

EDUCATION

Master of Science in Business Analytics

Aug 2023 – May 2025

Texas A&M University, Texas, USA

Bachelor of Business Administration (BBA)

Jain University, Bangalore, India